

Pedal settings: F1/F2 x 12 tonics

Cells: D-C || B-sep || E-F || G-A Per cell: top=flat, middle=natural, bottom=sharp * = enharmonic respelling

	C	G	D	A	E	B	F#	F	Bb	Eb	Ab	Db	Gb
F1 major													
Ionian0	????	????	????	????	????	????	????	????	????	????	????	????	????
Dorian0	????	????	????	????	????	????	????	????	????	????	????	????	????*
Phrygian0	????	????	????	????	????	????	????	????	????	????	????*	????*	????*
Lydian0	????	????	????	????	????	????	????	????	????	????	????	????	????
Mixolydian0	????	????	????	????	????	????	????	????	????	????	????	????	????
Aeolian0	????	????	????	????	????	????	????	????	????	????	????	????*	????*
Locrian0	????	????	????	????	????	????	????	????	????	????*	????*	????*	????*

F2 melodic minor

Ionian1	????	????	????	????	????	????	????	????*	????*	????*	????*	????*	????*
Dorian7	????	????	????	????	????	????	????	????	????	????	????	????	????*
Phrygian6	????	????	????	????	????	????	????	????	????	????	????*	????*	????*
Lydian5	????	????	????	????	????	????*	????*	????	????	????	????	????	????
Mixolydian4	????	????	????	????	????	????	????	????	????	????	????	????	????
Aeolian3	????	????	????	????	????	????	????	????	????	????	????	????*	????*
Locrian2	????	????	????	????	????	????	????	????	????	????*	????*	????*	????*

Mode reference (W=whole, H=half)

mode	pattern	char	tone
Lydian	W W W H W W H	#4	brightest major
Ionian	W W H W W W H	nat 7	major scale
Mixolydian	W W H W W H W	b7	dominant feel
Dorian	W H W W W H W	nat 6	bright minor
Aeolian	W H W W H W W	b6	natural minor
Phrygian	H W W W H W W	b2	Spanish color
Locrian	H W W H W W W	b2 b5	darkest

Families

F1	Ionian	major scale
F2	Dorian #7	melodic minor
F3	Aeolian #7	harmonic minor
F4	Ionian b6	harmonic major

Brightness

Lyd (brightest)
Ion
Mix
Dor
Aeo
Phr
Loc (darkest)

Pedal settings: F3/F4 x 12 tonics

Cells: D-C || B-sep || E-F || G-A Per cell: top=flat, middle=natural, bottom=sharp * = enharmonic respelling

	C	G	D	A	E	B	F#	F	Bb	Eb	Ab	Db	Gb
F3 harmonic minor													
Ionian5	????	????	????	????	????	????	* ???? *	????	????	????	????	????	????
Dorian4	????	????	????	????	????	????	????	????	????	????	????	????	???? *
Phrygian3	????	????	????	????	????	????	????	????	????	????	???? *	???? *	???? *
Lydian2	????	????	????	????	???? *	???? *	???? *	????	????	????	????	????	????
Mixolydian1	???? *	????	????	????	????	????	????	???? *	???? *	???? *	???? *	???? *	???? *
Aeolian7	????	????	????	????	????	????	????	????	????	????	????	???? *	???? *
Locrian6	????	????	????	????	????	????	????	????	????	???? *	???? *	???? *	???? *

F4 harmonic major

Ionian6	????	????	????	????	????	????	????	????	????	????	????	???? *	???? *
Dorian5	????	????	????	????	????	????	????	????	????	???? *	???? *	???? *	???? *
Phrygian4	????	????	????	????	????	????	????	???? *	???? *	???? *	???? *	???? *	???? *
Lydian3	????	????	????	????	????	????	????	????	????	????	????	????	???? *
Mixolydian2	????	????	????	????	????	????	????	????	????	????	???? *	???? *	???? *
Aeolian1	????	????	????	????	???? *	???? *	???? *	????	????	????	????	????	????
Locrian7	???? *	????	????	????	????	????	????	???? *	???? *	???? *	???? *	???? *	???? *

Improv tips per mode

Lydian	land on #4 as upper neighbor to 5; don't resolve #4→5
Ionian	use 7→1 leading-tone cadence; avoid resting on 4
Mixolydian	feature b7; cadence bVII→I (no leading tone)
Dorian	feature natural 6 (the modal marker); IV is major
Aeolian	stay diatonic; raising 7 drifts into harmonic minor
Phrygian	b2 is the signature; cadence bII→I; avoid V
Locrian	b2 and b5 as tension; unstable as tonic

Family alterations

(Ion-Dor-Phr-Lyd-
Mix-Aeo-Loc order)

F2: #1 #7 #6 #5 #4 #3 #2
 F3: #5 #4 #3 #2 #1 #7 #6
 F4: b6 b5 b4 b3 b2 b1 b7